Best-fit line (Biased random walk)

Given points (x_i, y_i) for $i \in [1, 2, \ldots, n]$, we aim to find the minimum sum of squared errors (SSE) between the points and the line $\hat{y} = cx$.

Let the error e_i be defined as the distance between the line and a given point (x_i, y_i).

$$\begin{aligned} e_i &= y_i - \hat{y}_i \\ &= y_i - cx_i \end{aligned}$$

The SSE (E) is then:

$$\begin{aligned} E &= \sum (e_i)^2 \\ &= \sum \left(y_i - cx_i\right)^2 \end{aligned}$$

The minimum occurs when the derivative of E is zero.

$$\begin{aligned} \frac{\mathrm{d}E}{\mathrm{d}c} = 0 &\implies \frac{\mathrm{d}}{\mathrm{d}c} \sum \left(y_i - cx_i\right)^2 = 0 \\ &\implies \sum 2\left(y_i - cx_i\right)(-x_i) = 0 \\ &\implies \sum \left(y_i - cx_i\right)(x_i) = 0 \\ &\implies \sum \left(x_i y_i - c\left(x_i^2\right)\right) = 0 \\ &\implies \sum \left(x_i y_i\right) - c \sum x_i^2 = 0 \\ &\implies c = \boxed{\frac{\sum \left(x_i \cdot y_i\right)}{\sum x_i^2}} \end{aligned}$$

Best-fit square root curve (Unbiased random walk)

Given points (x_i, y_i) for $i \in [1, 2, \ldots, n]$, we aim to find the minimum sum of squared errors (SSE) between the points and the curve $\hat{y} = c\sqrt{x}$.

Let the error e_i be defined as the distance between the curve and a given point (x_i, y_i).

$$\begin{aligned} e_i &= y_i - \hat{y}_i \\ &= y_i - c\sqrt{x_i} \end{aligned}$$

The SSE (E) is then:

$$\begin{aligned} E &= \sum (e_i)^2 \\ &= \sum \left(y_i - c\sqrt{x_i}\right)^2 \end{aligned}$$

The minimum occurs when the derivative of E is zero.

$$\begin{aligned} \frac{\mathrm{d}E}{\mathrm{d}c} = 0 &\implies \frac{\mathrm{d}}{\mathrm{d}c} \sum \left(y_i - c\sqrt{x_i}\right)^2 = 0 \\ &\implies \sum 2\left(y_i - c\sqrt{x_i}\right)\left(-\sqrt{x_i}\right) = 0 \end{aligned}$$

$$\implies \sum \left(y_i - c\sqrt{x_i}\right)\left(\sqrt{x_i}\right) = 0$$
$$\implies \sum \left(\sqrt{x_i} \cdot y_i\right) - c \sum x_i = 0$$
$$\implies c = \boxed{\frac{\sum \left(\sqrt{x_i} \cdot y_i\right)}{\sum x_i}}$$